

Modeling Loci and Intersections with Conic Sections

Building a circle from a distance condition, using the focus and directrix of a parabola, and solving conic intersections

Directions

- Problems 1–3 are a warm-up, one on each idea. After that the three are mixed, so read each problem and decide which tool it wants before you start.
- For a locus problem, translate the sentence into a distance equation first, then simplify.
- For an intersection, substitute one equation into the other and report every point, not just the first.

1. A point $P(x, y)$ moves so that its distance from $C(3, -2)$ is always 5. Write the equation of its path in the expanded form $x^2 + y^2 + Dx + Ey + F = 0$.

Answer: _____

2. A point is equidistant from the focus $(0, 3)$ and the directrix $y = -3$; its path is the parabola $x^2 = 12y$. Find y when $x = 6$.

Answer: _____

3. Find the x -coordinates where the parabola $y = x^2$ meets the horizontal line $y = 4$.

Answer: _____

4. A circle has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. Complete the square in x and in y , then state the radius.

Answer: _____

5. A parabola has its vertex at the origin and its focus at $(0, 5)$, so its equation has the form $x^2 = 4py$.

(a) Find the value of $4p$.

Answer: _____

(b) Find both x -values on the parabola when $y = 5$.

Answer: _____

6. Find every intersection point of the parabola $y = x^2 - 3$ and the line $y = 2x$. Give your answers as ordered pairs.

Answer: _____

7. A point $P(x, y)$ moves so that its distance from $A(0, 0)$ is always twice its distance from $B(6, 0)$.

- (a) Square both distances, set up $PA^2 = 4PB^2$, and simplify to the form $x^2 + y^2 + Dx + Ey + F = 0$.

Answer: _____

- (b) Complete the square to find the radius of the resulting circle.

Answer: _____

8. Find every intersection point of the circle $x^2 + y^2 = 25$ and the line $y = x + 1$. Give your answers as ordered pairs.

Answer: _____

9. A parabola is the set of points equidistant from the focus $(2, 0)$ and the directrix $x = -2$. Its equation is $y^2 = 8x$.

(a) Find both y -values on the parabola when $x = 2$.

Answer: _____

(b) Explain, using the equidistance definition rather than the equation, why $(2, 4)$ lies on this parabola.

Answer: _____

10. A point $P(x, y)$ moves so that the sum of the squares of its distances from $(-3, 0)$ and $(3, 0)$ is always 50.

(a) Set up $PA^2 + PB^2 = 50$ and simplify to the form $x^2 + y^2 + F = 0$.

Answer: _____

(b) Describe the locus: what shape is it, where is it centred, and how big is it? Justify each part from your equation.

Answer: _____

11. Find every intersection point of the circle $x^2 + y^2 = 25$ and the parabola $y = x^2 - 13$. Give your answers as ordered pairs. (There are four.)

Answer: _____

12. The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point.

- (a) Substitute the line into the circle, set the discriminant of the resulting quadratic to zero, and find both values of k .

Answer: _____

- (b) Explain what “exactly one point” means about the quadratic’s roots and what it means about the picture.

Answer: _____